

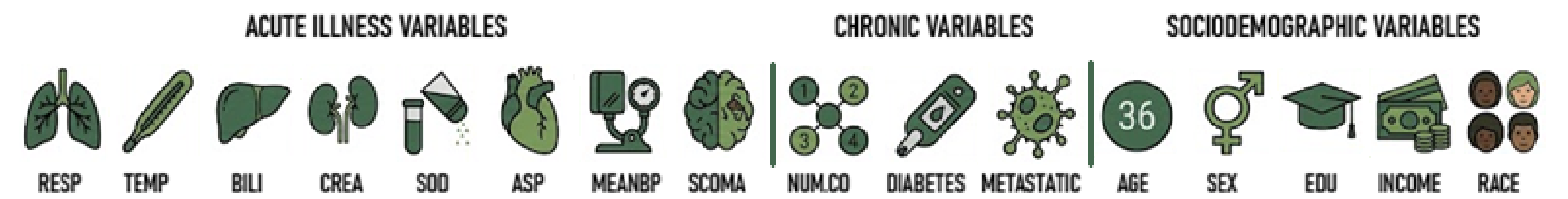
Beyond Acute Biomarkers: Predicting Two-Month Functional Disability in ICU Oncologic Patients Using Frailty Embeddings of Chronic and Sociodemographic Features

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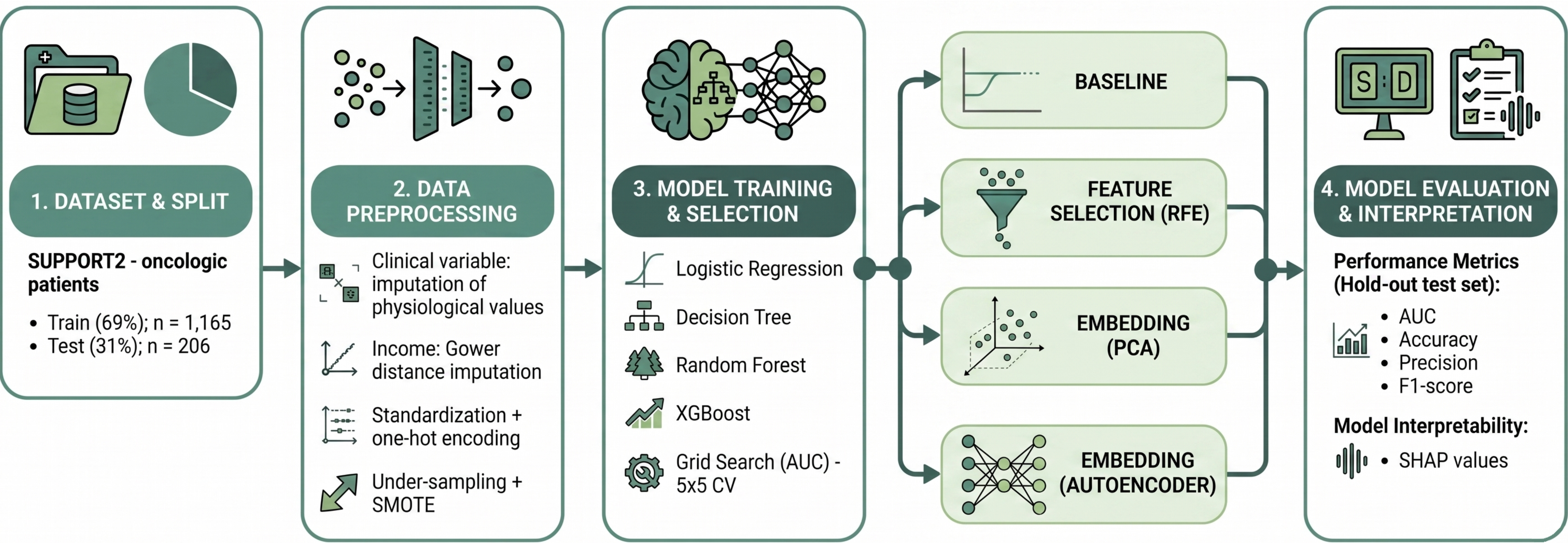
INTRODUCTION

Oncologic patients admitted to the Intensive Care Unit (ICU) represent a clinically complex population, where outcomes are shaped not only by acute illness severity but also by pre-existing chronic conditions and sociodemographic background. However, the prognostic value of these chronic and sociodemographic factors is often overlooked in favor of acute clinical measurements [1]. This study investigates whether **latent frailty representations** – obtained by embedding chronic and sociodemographic features – can predict functional disability at two months from ICU admission (sfdm2). The outcome was defined as a binary variable: **fully independent** (0) or **functionally impaired** (1), excluding patients who died during the observation period. Data were drawn from the SUPPORT2 dataset, a large prospective study of seriously ill hospitalized patients.

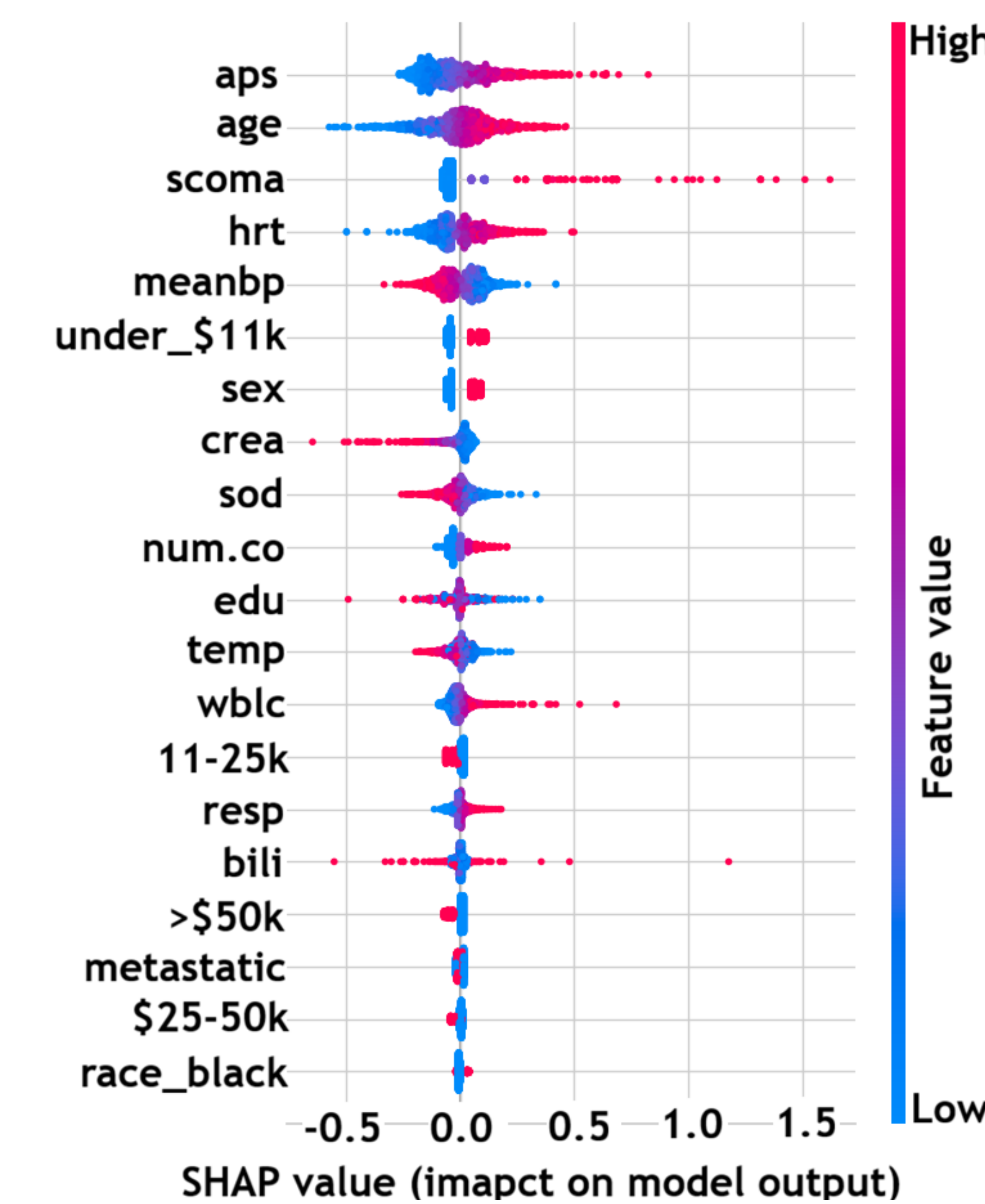


METHODS

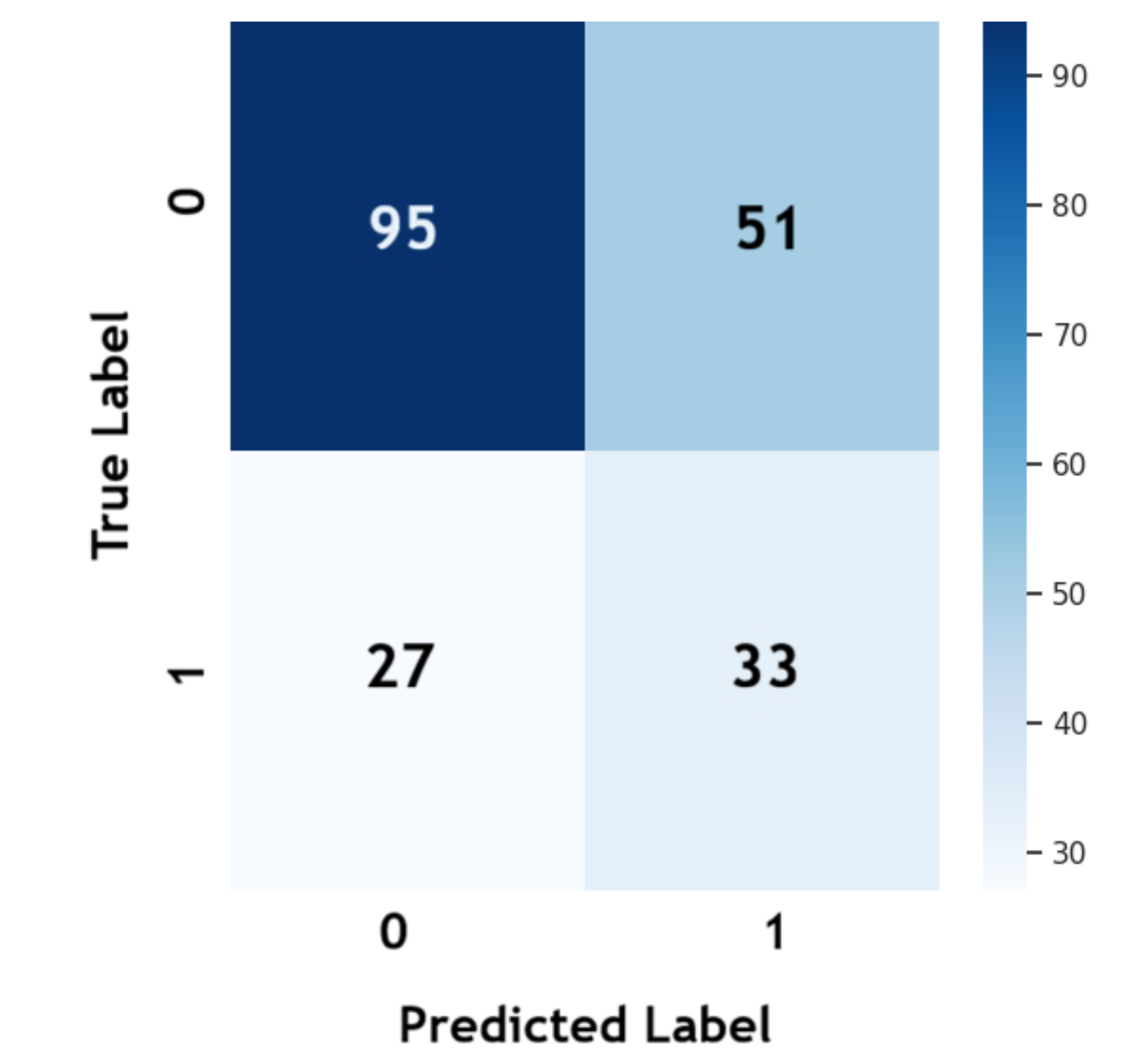
We analyzed a cohort of oncologic patients from the SUPPORT2 dataset to predict a binary functional disability outcome, partitioned into a training set (**n = 1165**) and a hold-out test set (**n = 206**). All preprocessing and resampling steps were embedded within a nested stratified **cross-validation** framework (5 inner and 5 outer folds) to prevent data leakage. Missing physiological variables were imputed using clinical constants from the original SUPPORT investigators [2]; missing income data was imputed via **Gower** distance matching. Continuous features were **standardized**, categorical features one-hot encoded, and class imbalance addressed through a hybrid **random under-sampling** and **SMOTE** approach. **Four classifiers** were evaluated (Logistic Regression, Decision Tree, Random Forest, XGBoost) with hyperparameters tuned via Grid Search on AUC. We compared four feature strategies: (1) all features as **baseline**; (2) Recursive Feature Elimination (**RFE**); (3) **PCA** reduction of chronic and sociodemographic variables into 3 components (Exp. Var. = 0.69); and (4) Three **autoencoder-derived** latent embeddings of the same variables – both concatenated with acute features. Final performance was assessed on the hold-out set via **AUC**, accuracy, precision, and F1-score, with **SHAP** values for interpretability.



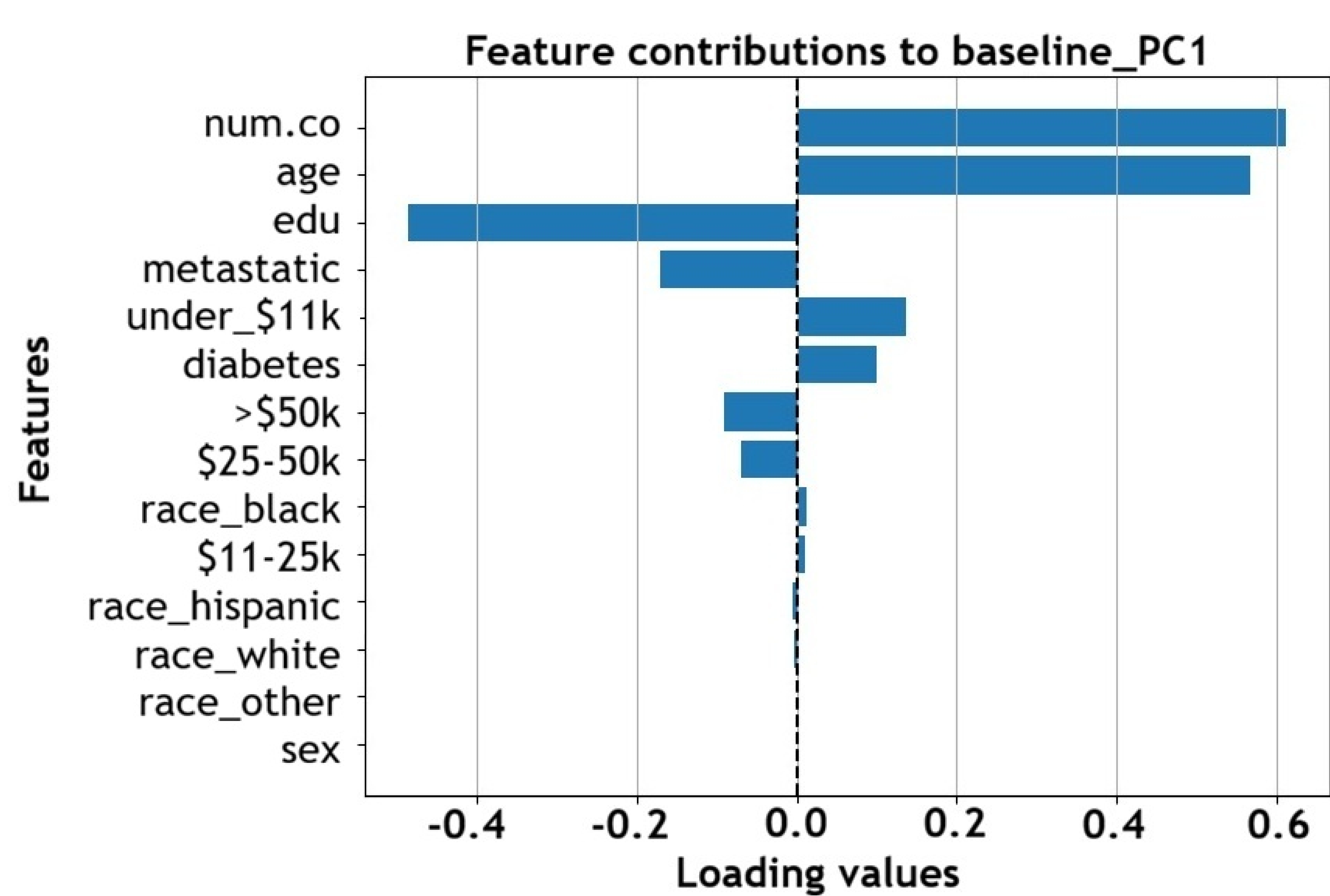
RESULTS



SHAP Values of Logistic Regression from Baseline Pipeline.



Confusion matrix of the Autoencoder Pipeline inference on the held-out test set.



Features contribution to PC1 computation in PCA Pipeline.

		CV (train)	CV (validation)	Held out (test)
Baseline	AUC (±std)	0.668±0.008	0.639±0.030	0.6244
	Accuracy (±std)	0.616±0.021	0.606±0.028	0.6165
RFE Pipeline	AUC (±std)	0.657±0.007	0.645±0.030	0.6039
	Accuracy (±std)	0.627±0.046	0.591±0.045	0.5534
PCA Pipeline	AUC (±std)	0.657±0.009	0.626±0.025	0.6084
	Accuracy (±std)	0.603±0.024	0.583±0.041	0.6117
AE Pipeline	AUC (±std)	0.660±0.008	0.634±0.027	0.6156
	Accuracy (±std)	0.602±0.025	0.585±0.036	0.6214

Results table of the best model of all pipelines (Logistic Regression) in terms of AUC and Accuracy for Train and Test splits of the CV and Held out Test Set.

CONCLUSIONS

Autoencoder + Logistic Regression achieved the best balance of performance, stability, and generalization, with similar discrimination across models. Strong dimensionality reduction did not reduce performance, suggesting that a compressed latent representation can preserve most of the predictive signal in the original feature space.

LIMITATIONS

- Binary outcome (operator-dependent, unbalanced, information loss)
- Autoencoder overkill for low-dimensional feature space
- Reduced interpretability of deep latent models
- Loss of features due to missingness

REFERENCES

[1] Bosch-Compte R. et al., Oncology Letters, vol. 26, no. 6, 2023; [2] Knaus WA et al., Ann Intern Med. 1995 Feb 1;122(3):191-203.



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